LOK JAGRUTI KENDRA UNIVERSITY

Syllabus for two Years
School of Computer Applications,
Master of Computer Applications (MCA)
Semester - I

Course Code	**240110102**			
Category	**Core Subject**			
Course Title	**Relational Database Management System (RDBMS)**			
Scheme and Credits	**Theory**	**Tutorial**	**Lab**	**Credits**
	4	**0**	**0**	**4**
Pre-requisites (if any)	Basic knowledge of data storing and retrieving mechanism in computer and types of software and applications using different types of data to be stored and retrieve as per the need. Proficiency in any programming language. The course teaching language is English, so students have to have communication, reading and apprehension skills of English.			

1. **Course Objectives:**

1	To understand the fundamental concepts of database systems, their architecture, types, and languages, and to differentiate between traditional file systems and modern DBMS/RDBMS approaches including an overview of NoSQL and NewSQL technologies.
2	To develop the ability to design and interpret Entity-Relationship (ER) models and effectively convert them into relational schemas for real-world applications, ensuring a strong foundation in logical database design.
3	To understand the principles of relational schema design, apply normalization techniques using functional dependencies, and ensure data integrity by eliminating anomalies through proper constraints and Codd's rules.
4	To understand various file organization and indexing techniques, and to analyze query execution strategies through cost estimation and optimization methods for efficient database performance.
5	To gain knowledge of transaction management concepts, including ACID properties, concurrency control, and recovery techniques, ensuring consistency and reliability in multi-user database environments.

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2. Course Contents

Unit	Course Content	Weightage
Unit I	**Introduction to Database System (CO1)** Data, Information, File-based Data Management, Difference between file-system and database management system, Differentiate between Database and Database Management System, Concept of Data dictionary in DBMS **Database**: Users, Data Models, Data abstraction, ANSI/SPARC Architecture, Logical and Physical data independence, DBMS Languages: DDL, DML, DCL (definitions only) **RDBMS**: Examples of RDBMS, Classifications of Database Management Systems – Relational and Non-Relational Database, Benefits of using RDBMS, Introduction to NoSQL and NewSQL (overview only)	**20%**
Unit II	**Entity Relationship Modelling (CO2)** **Components of an E-R Model:** Entity, Entity Sets, Attributes and keys – Super Key, Primary Key, Candidate Key, Alternate Key, Relationship Types, Composite entities, Weak entity Relational Database design by ER to Relational Mapping **Examples for ER Diagram:** College Management System, Railway Reservation System, Hospital Management System, Library Management System, e-Business Management System (can take any eBusiness example like online shopping, online food ordering etc.…)	**20%**